

Technical Project Report

Signal Processing and Blind Source Separation

Independent Component Analysis

**Foundations, algorithms, applications, and a reproducible audio
demonstration**

Hamza LOUKILI

August 2026

github.com/loukili27/ICA-Voices

Abstract

Signals encountered in audio, neuroscience, biomedical sensing, telecommunications, and finance are often mixtures of several latent phenomena. Independent Component Analysis (ICA) is a statistical framework for estimating latent components that are as independent as possible from multivariate observations. This report presents the linear ICA model, its identifiability assumptions, the principal estimation criteria, the FastICA algorithm, and representative application domains. A reproducible audio experiment then applies FastICA to three mono mixtures containing 264,515 samples each at 44.1 kHz. The algorithm converges in four iterations and reconstructs the observations with a relative root mean square error of 4.219×10^{-16} . Because isolated reference sources are unavailable, these measurements establish numerical consistency rather than perfect source recovery. A brief final section positions ICA relative to Principal Component Analysis (PCA) without making that comparison the central subject of the report.

Contents

1	Introduction	1
2	Theoretical foundation of ICA	1
2.1	Linear instantaneous mixing model	1
2.2	Statistical independence	3
2.3	Non-Gaussianity and identifiability	3
2.4	Intrinsic ambiguities	3
3	Mathematical criteria and estimation	3
3.1	Centering and whitening	3
3.2	Kurtosis	4
3.3	Negentropy	4
3.4	Mutual information	4
3.5	Maximum likelihood	5
3.6	FastICA fixed-point update	5
4	Applications and use cases	5
5	Reproducible audio demonstration	6
5.1	Input data and preprocessing	6
5.2	FastICA configuration	6
6	Experimental results	8
6.1	Dependence in the observations	8
6.2	Estimated independent components	8
6.3	Numerical validation	8
7	Limitations and responsible interpretation	9
8	Practical positioning relative to PCA	10
9	Conclusion	10

List of Figures

1	Linear instantaneous ICA model. The unknown matrix $\mathbf{A}$ mixes the latent sources, while ICA estimates an unmixing matrix $\mathbf{W}$ from the observations. . .	2
2	Intuitive blind source separation example retained from the original report. .	2
3	Observed audio mixtures used by the demonstration: three mono signals sampled at 44.1 kHz over 5.998 seconds.	7
4	Three components estimated by FastICA. Their order, sign, and amplitude are not directly comparable with unknown source labels because of ICA's intrinsic ambiguities.	9

1 Introduction

Signals are ubiquitous and frequently overlap. Voices recorded in a room, readings collected by multiple sensors, or electrical potentials measured on the scalp can each contain contributions from several underlying sources. Separating those sources is essential when the mixtures obscure the information of interest.

Blind source separation seeks to recover latent signals while knowing little or nothing about the sources or the physical mixing process. Independent Component Analysis (ICA) provides a principled solution when the observations can be modeled as linear mixtures of statistically independent, mainly non-Gaussian sources [1, 3]. Instead of relying only on covariance, ICA exploits higher-order and information-theoretic structure.

ICA is not a single-purpose audio technique. It is a general latent-variable method whose usefulness depends on the physical model, the statistical independence assumption, source non-Gaussianity, and the availability of sufficiently distinct observations.

This report develops the ICA model and its assumptions, derives the main estimation criteria, explains FastICA, reviews representative applications, and presents a fully reproducible audio demonstration. Only after the ICA analysis is complete is PCA mentioned briefly to clarify when each method is appropriate.

2 Theoretical foundation of ICA

2.1 Linear instantaneous mixing model

Let the unknown source vector be

$$\mathbf{s}(t) = [s_1(t), s_2(t), \dots, s_n(t)]^\top, \quad (1)$$

and let the observed mixtures be

$$\mathbf{x}(t) = [x_1(t), x_2(t), \dots, x_m(t)]^\top. \quad (2)$$

The basic ICA model assumes an instantaneous linear mixture

$$\mathbf{x}(t) = \mathbf{A}\mathbf{s}(t), \quad (3)$$

where $\mathbf{A} \in \mathbb{R}^{m \times n}$ is an unknown mixing matrix. Each coefficient a_{ij} represents the contribution of source j to observation i .

In the determined setting, $m = n$ and $\mathbf{A}$ is invertible. ICA estimates an unmixing matrix $\mathbf{W}$ so that

$$\mathbf{y}(t) = \mathbf{W}\mathbf{x}(t) = \mathbf{W}\mathbf{A}\mathbf{s}(t) \approx \mathbf{s}(t). \quad (4)$$

Figure 2 retains the intuitive illustration from the original report: several physical sources contribute simultaneously to each microphone signal, and ICA attempts to recover one latent component per output. It complements the compact matrix representation in Figure 1.

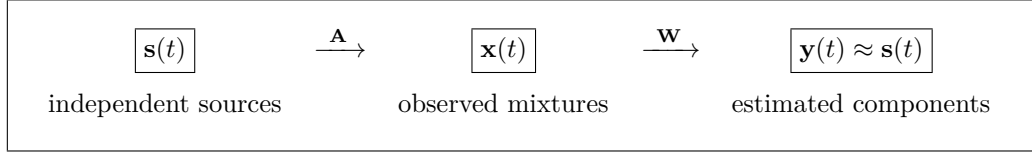


Figure 1: Linear instantaneous ICA model. The unknown matrix $\mathbf{A}$ mixes the latent sources, while ICA estimates an unmixing matrix $\mathbf{W}$ from the observations.

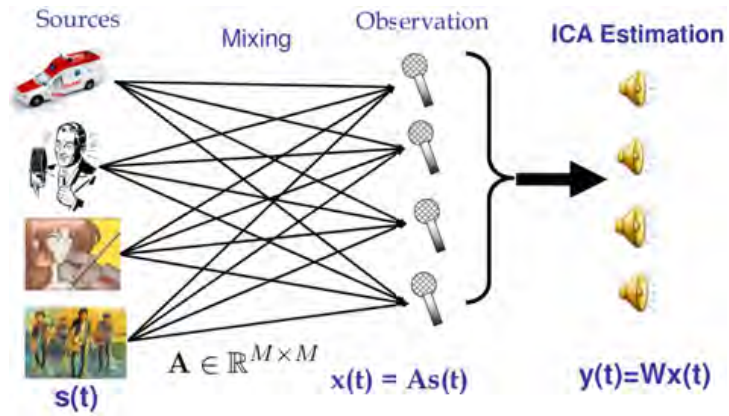


Figure 2: Intuitive blind source separation example retained from the original project materials. The source signals $\mathbf{s}(t)$ are mixed into microphone observations $\mathbf{x}(t) = \mathbf{A}\mathbf{s}(t)$, before ICA estimates $\mathbf{y}(t) = \mathbf{W}\mathbf{x}(t)$. The original external source of this legacy illustration should be confirmed before public redistribution.

2.2 Statistical independence

The central assumption is that the sources are mutually independent. For all admissible values,

$$p_{s_1, \dots, s_n}(s_1, \dots, s_n) = \prod_{i=1}^n p_{s_i}(s_i). \quad (5)$$

Independence implies zero covariance when moments exist, but the converse is generally false. Two variables can be uncorrelated while remaining dependent through nonlinear or higher-order relationships. This is precisely why diagonalizing a covariance matrix with PCA does not, in general, solve the ICA problem.

2.3 Non-Gaussianity and identifiability

The classical linear ICA model permits at most one Gaussian source. A multivariate Gaussian distribution that has been whitened is rotationally symmetric: any orthogonal rotation produces the same distribution. Therefore covariance and Gaussian likelihood alone cannot identify a unique rotation. Non-Gaussian source distributions break this rotational ambiguity and make the independent directions identifiable, apart from unavoidable permutation and scaling ambiguities [4].

The central-limit intuition is that a normalized sum of independent random variables tends to be more Gaussian than the original variables. ICA algorithms can therefore search for projections that are maximally non-Gaussian.

2.4 Intrinsic ambiguities

Even under ideal assumptions, ICA cannot determine the original component order because

$$\mathbf{A}\mathbf{s} = (\mathbf{A}\mathbf{P}^{-1})(\mathbf{P}\mathbf{s}) \quad (6)$$

for any permutation matrix $\mathbf{P}$. It also cannot determine component scale because a source may be multiplied by a nonzero scalar if the corresponding column of $\mathbf{A}$ is divided by the same scalar. Consequently, sign changes, scale changes, and component permutations do not indicate failure.

3 Mathematical criteria and estimation

3.1 Centering and whitening

ICA is usually applied after centering and whitening. If $\boldsymbol{\mu} = \mathbb{E}[\mathbf{x}]$, centering gives

$$\mathbf{x}_c = \mathbf{x} - \boldsymbol{\mu}. \quad (7)$$

Let the covariance eigendecomposition be

$$\mathbf{C}_x = \mathbb{E}[\mathbf{x}_c \mathbf{x}_c^\top] = \mathbf{E} \mathbf{D} \mathbf{E}^\top. \quad (8)$$

A common whitening transform based on this eigendecomposition is

$$\mathbf{z} = \mathbf{D}^{-1/2} \mathbf{E}^\top \mathbf{x}_c, \quad \mathbb{E}[\mathbf{z} \mathbf{z}^\top] = \mathbf{I}. \quad (9)$$

Whitening removes second-order correlations and constrains the remaining ICA mixing transform to a rotation. It simplifies the optimization but does not by itself establish independence.

3.2 Kurtosis

For a zero-mean variable y , the fourth-order cumulant is

$$\text{kurt}(y) = \mathbb{E}[y^4] - 3(\mathbb{E}[y^2])^2. \quad (10)$$

A Gaussian variable has zero excess kurtosis. Positive and negative values characterize super-Gaussian and sub-Gaussian behavior, respectively. Maximizing the absolute kurtosis of standardized projections is one possible ICA contrast, although kurtosis can be sensitive to outliers.

3.3 Negentropy

Differential entropy is

$$H(y) = - \int p_y(u) \log p_y(u) du. \quad (11)$$

Among variables with equal variance, the Gaussian distribution has maximum entropy. Negentropy therefore defines a nonnegative measure of non-Gaussianity:

$$J(y) = H(y_{\text{Gauss}}) - H(y) \geq 0. \quad (12)$$

Because exact entropy estimation is difficult, FastICA uses robust approximations based on smooth nonlinear functions such as $G(u) = \log \cosh(u)$ [2].

3.4 Mutual information

Mutual information measures dependence between the components of $\mathbf{y}$:

$$I(y_1, \dots, y_n) = \sum_{i=1}^n H(y_i) - H(\mathbf{y}) \geq 0. \quad (13)$$

It equals zero if and only if the components are mutually independent under standard regularity conditions. For whitened data, minimizing mutual information is closely connected to maximizing component non-Gaussianity [1].

3.5 Maximum likelihood

If $\mathbf{y} = \mathbf{W}\mathbf{x}$ and the source densities p_i are modeled, the change-of-variables formula gives

$$p_{\mathbf{x}}(\mathbf{x}; \mathbf{W}) = |\det \mathbf{W}| \prod_{i=1}^n p_i(\mathbf{w}_i^T \mathbf{x}). \quad (14)$$

For T samples, the log-likelihood is

$$\ell(\mathbf{W}) = T \log |\det \mathbf{W}| + \sum_{t=1}^T \sum_{i=1}^n \log p_i(\mathbf{w}_i^T \mathbf{x}(t)). \quad (15)$$

Maximizing this expression encourages an invertible transform whose outputs follow factorized source models.

3.6 FastICA fixed-point update

For whitened samples $\mathbf{z}$ and a unit vector $\mathbf{w}$, one FastICA update has the form

$$\mathbf{w}^+ = \mathbb{E}[\mathbf{z} g(\mathbf{w}^T \mathbf{z})] - \mathbb{E}[g'(\mathbf{w}^T \mathbf{z})] \mathbf{w}, \quad \mathbf{w} \leftarrow \frac{\mathbf{w}^+}{\|\mathbf{w}^+\|_2}. \quad (16)$$

The parallel algorithm updates all rows of $\mathbf{W}$ and then applies symmetric decorrelation,

$$\mathbf{W} \leftarrow (\mathbf{W}\mathbf{W}^T)^{-1/2} \mathbf{W}, \quad (17)$$

until the directions stabilize. The fixed-point approach is fast and avoids manually selecting a learning rate [2].

4 Applications and use cases

ICA is most valuable when several recorded variables contain overlapping contributions from latent processes and those processes have a plausible interpretation. Its role varies across domains:

- **Audio and acoustics:** separating simultaneously recorded voices or instruments, enhancing speech, analyzing microphone arrays, and studying the cocktail-party problem.
- **Neuroscience:** removing eye-blink, muscle, cardiac, and line-noise artifacts from EEG or MEG; ICA can also provide an exploratory decomposition of brain activity [6].
- **Biomedical sensing:** isolating physiological components and measurement artifacts in ECG, imaging, and multimodal sensor recordings.
- **Images and spatial data:** decomposing image collections into latent spatial patterns and extracting statistically distinct features.
- **Telecommunications:** blind channel identification, multi-user detection, and source separation in multi-antenna systems.

- **Industrial monitoring:** separating process variations, disturbances, and sensor effects for exploratory fault detection.
- **Finance:** exploring statistically independent factors in correlated returns, subject to strong cautions about nonstationarity and economic interpretation.

These examples share a common objective: recovering interpretable latent mechanisms rather than merely compressing the observations. ICA is therefore especially relevant when the components themselves matter.

The assumptions must always be checked against the application. Reverberant audio, propagation delays, nonlinear sensors, nonstationary sources, and time-varying mixtures require extensions beyond instantaneous linear ICA. When the latent sources are not plausibly independent, an ICA decomposition may still be mathematically computable but scientifically misleading.

5 Reproducible audio demonstration

5.1 Input data and preprocessing

The repository contains three precomputed mono WAV mixtures: `ICA mix 1.wav`, `ICA mix 2.wav`, and `ICA mix 3.wav`. Each is a 16-bit PCM signal sampled at 44,100 Hz. The shortest common segment contains 264,515 samples, corresponding to 5.998 seconds. The $264,515 \times 3$ observation matrix has numerical rank three.

The notebook converts integer samples to floating point values, aligns all observations to the same length, and centers each channel. Three genuinely distinct observations are essential: duplicating one recording would produce a rank-deficient matrix and would not create enough information to recover three components.

5.2 FastICA configuration

The implementation uses scikit-learn’s **FastICA** estimator with the parameters in Table 1 [7]. The random state is fixed for reproducibility. Unit-variance whitening is specified explicitly, and the `logcosh` contrast provides a smooth negentropy approximation.

After estimation, each component is normalized independently to 95% of the signed 16-bit range before WAV export. This prevents clipping and integer overflow; it replaces the arbitrary volume multiplier used in the historical notebook.

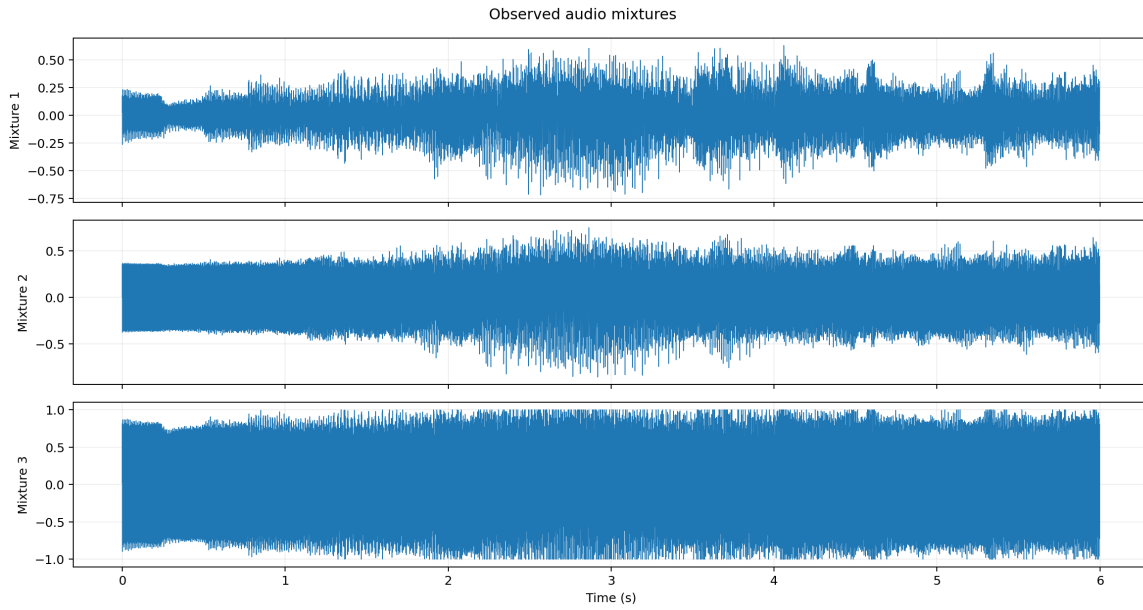


Figure 3: Observed audio mixtures used by the demonstration: three mono signals sampled at 44.1 kHz over 5.998 seconds.

Table 1: FastICA configuration used in the notebook.

Parameter	Value
Number of components	3
Algorithm	Parallel
Whitening	Unit variance
Nonlinearity	logcosh
Maximum iterations	2,000
Tolerance	10^{-5}
Random state	42

6 Experimental results

6.1 Dependence in the observations

The observed mixtures are strongly correlated:

$$\mathbf{R}_X = \begin{bmatrix} 1.000 & 0.846 & 0.784 \\ 0.846 & 1.000 & 0.972 \\ 0.784 & 0.972 & 1.000 \end{bmatrix}. \quad (18)$$

The large off-diagonal values show that the observations share substantial linear structure.

6.2 Estimated independent components

FastICA converged in four iterations. The estimated components have an identity sample correlation matrix to the displayed precision. Their Fisher kurtosis values are

$$[1.401, -1.499, 2.775], \quad (19)$$

so all three components depart from Gaussianity at the fourth-order level. The signs indicate different distribution shapes: positive excess kurtosis is associated with heavier tails, while negative excess kurtosis indicates a flatter distribution. These values support the non-Gaussianity used by FastICA, but they are not by themselves proof that the recovered components match the unknown physical sources.

6.3 Numerical validation

Table 2: Measured outputs of the reproducible experiment.

Metric	Result
FastICA iterations	4
Largest absolute off-diagonal ICA correlation	$< 10^{-4}$
Reconstruction RMSE	1.608×10^{-16}
Relative reconstruction RMSE	4.219×10^{-16}
Exported separated WAV files	3

The near-machine-precision reconstruction error confirms that the estimated mixing and unmixing transforms consistently factorize the same observation matrix. It is not a source-separation score: inverse transformation reconstructs the mixtures used for fitting.

The exported WAV files are stored in `outputs/separated/` and can be played inside the notebook. Listening provides a qualitative check, but a rigorous SDR or SI-SDR evaluation would require the isolated reference sources.

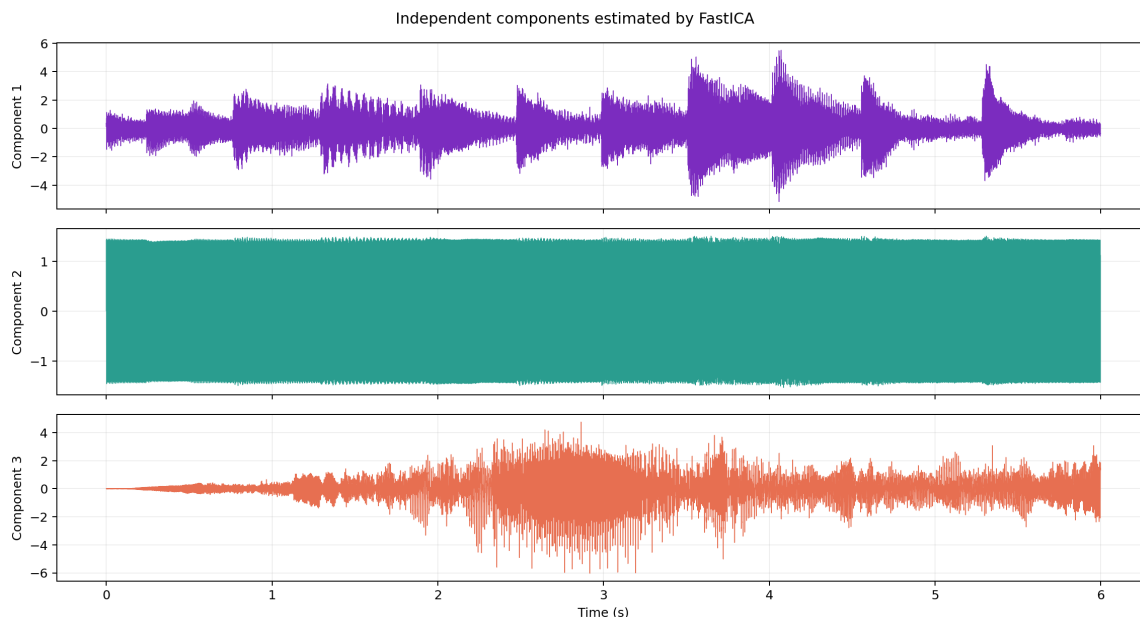


Figure 4: Three components estimated by FastICA. Their order, sign, and amplitude are not directly comparable with unknown source labels because of ICA’s intrinsic ambiguities.

7 Limitations and responsible interpretation

The experiment supports numerical convergence, decorrelation, non-Gaussian component estimation, reconstruction consistency, and qualitative listening. It does not support a claim of objectively perfect recovery because:

- the isolated reference sources and original mixing matrix are unavailable;
- zero correlation is weaker than statistical independence;
- the instantaneous model does not explicitly represent echo, propagation delay, nonlinear mixing, or a time-varying room;
- the method requires sufficiently distinct observations and a full-rank mixture;
- multiple Gaussian sources are not identifiable without additional structure.

These qualifications correct the stronger wording in the first report version while preserving its central theoretical motivation.

8 Practical positioning relative to PCA

PCA is mentioned here only to prevent a common methodological confusion. PCA and ICA are both linear transformations, but they are designed for different goals [5]. PCA is primarily a representation method: it orders orthogonal directions by explained variance. ICA is primarily a source-separation method: it searches for latent directions that are as statistically independent as possible.

Table 3: Practical distinction between PCA and ICA.

Question	PCA	ICA
Main objective	Preserve variance in fewer orthogonal variables	Recover statistically independent latent components
Main information	Covariance and second-order structure	Non-Gaussianity and higher-order structure
Choose it for	Dimension reduction, visualization, compression, denoising	Blind source separation, artifact removal, latent-process discovery
Component order	Ranked by explained variance	No intrinsic ranking
Main caution	High variance is not necessarily meaningful	Independence and mixing assumptions must be plausible

Use PCA when the objective is to obtain a compact representation of high-dimensional data. Use ICA when the observations are believed to be mixtures and the latent signals themselves are the target. The methods can also be complementary: PCA may reduce dimension and whiten the data before ICA estimates an independent rotation.

9 Conclusion

ICA is a fundamental method for discovering statistically independent structure in multivariate data. Its usefulness extends from the cocktail-party problem to neuroscience, biomedical sensing, imaging, telecommunications, industrial monitoring, and exploratory factor analysis. The method relies on a suitable mixture model, source independence, non-Gaussianity, and sufficiently informative observations.

The audio experiment makes the complete workflow concrete: align and center the mixtures, whiten the data, estimate the unmixing transform with FastICA, export the components, and validate the numerical factorization. The results establish convergence and internal consistency while the absence of isolated references prevents a claim of objectively perfect separation.

The complete reproducible implementation, input mixtures, and separated outputs are available at

github.com/loukili27/ICA-Voices.

References

- [1] P. Comon, “Independent component analysis, a new concept?” *Signal Processing*, vol. 36, no. 3, pp. 287-314, 1994. DOI: 10.1016/0165-1684(94)90029-9.
- [2] A. Hyvärinen, “Fast and robust fixed-point algorithms for independent component analysis,” *IEEE Transactions on Neural Networks*, vol. 10, no. 3, pp. 626-634, 1999. DOI: 10.1109/72.761722.
- [3] A. Hyvärinen and E. Oja, “Independent component analysis: algorithms and applications,” *Neural Networks*, vol. 13, no. 4-5, pp. 411-430, 2000. DOI: 10.1016/S0893-6080(00)00026-5.
- [4] A. Hyvärinen, J. Karhunen, and E. Oja, *Independent Component Analysis*. New York: Wiley, 2001.
- [5] I. T. Jolliffe, *Principal Component Analysis*, 2nd ed. New York: Springer, 2002.
- [6] S. Makeig, A. J. Bell, T.-P. Jung, and T. J. Sejnowski, “Independent component analysis of electroencephalographic data,” in *Advances in Neural Information Processing Systems* 8, 1996, pp. 145-151.
- [7] Scikit-learn developers, “FastICA API documentation,” [scikit-learn.org](https://scikit-learn.org/stable/modules/generated/sklearn.decomposition.FastICA.html), accessed August 24, 2026.